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(54) **REFRIGERANT ISOLATION FOR COOLING SOLUTIONS**

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(57)

ABSTRACT

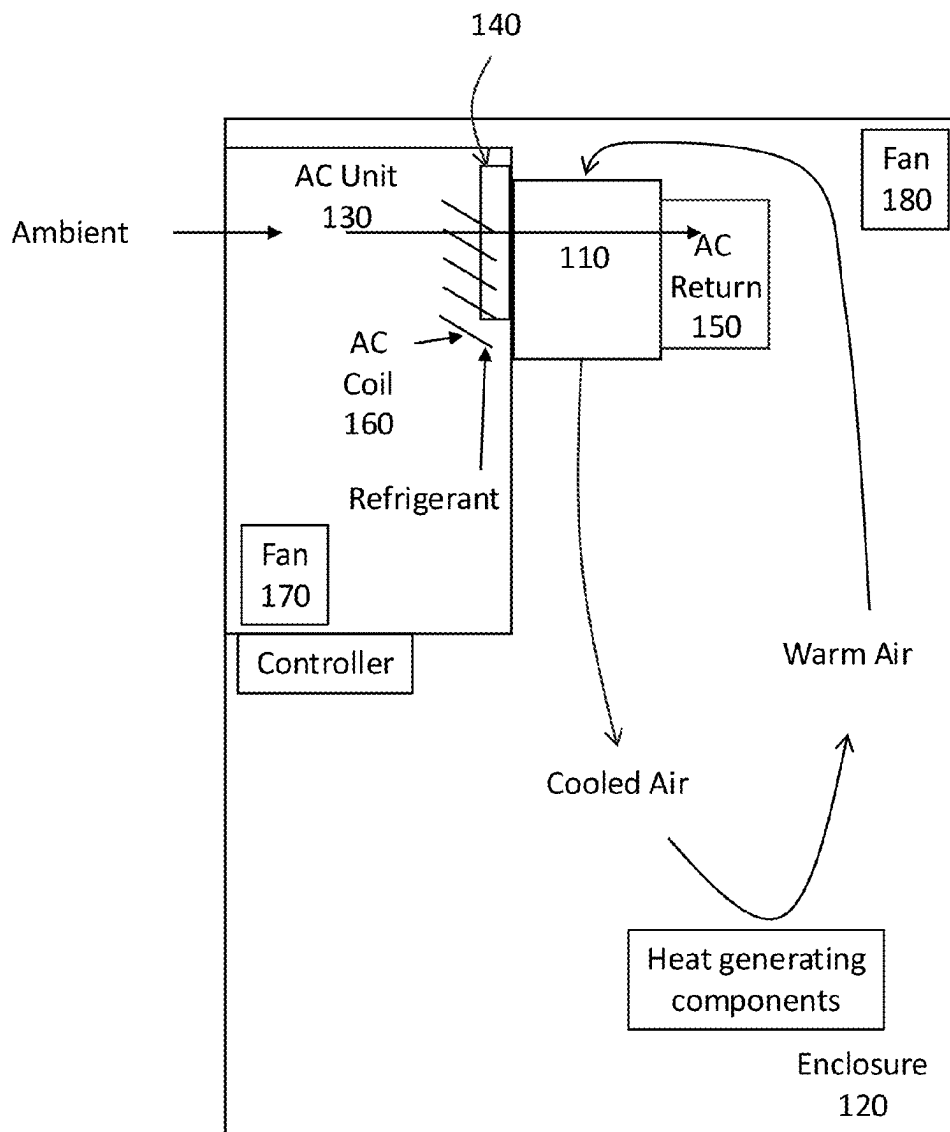
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An isolation core is used to isolate an air conditioning (AC) unit and an external environment such as an enclosure to be cooled. An isolation core may take the form of a cross-flow or counterflow heat exchanger which acts to thermally couple the flows of an AC unit and an enclosure air inlet and return without permitting the flows to intermix. The isolation core thus facilitates cooling of enclosures while isolating flammable refrigerant from the enclosure air, requiring the refrigerant to remain within the AC unit even if an interior component of the AC unit such as a cooling coil, etc., leaks.

(22) Filed: **Mar. 6, 2025**

Related U.S. Application Data

(60) Provisional application No. 63/562,511, filed on Mar. 7, 2024.



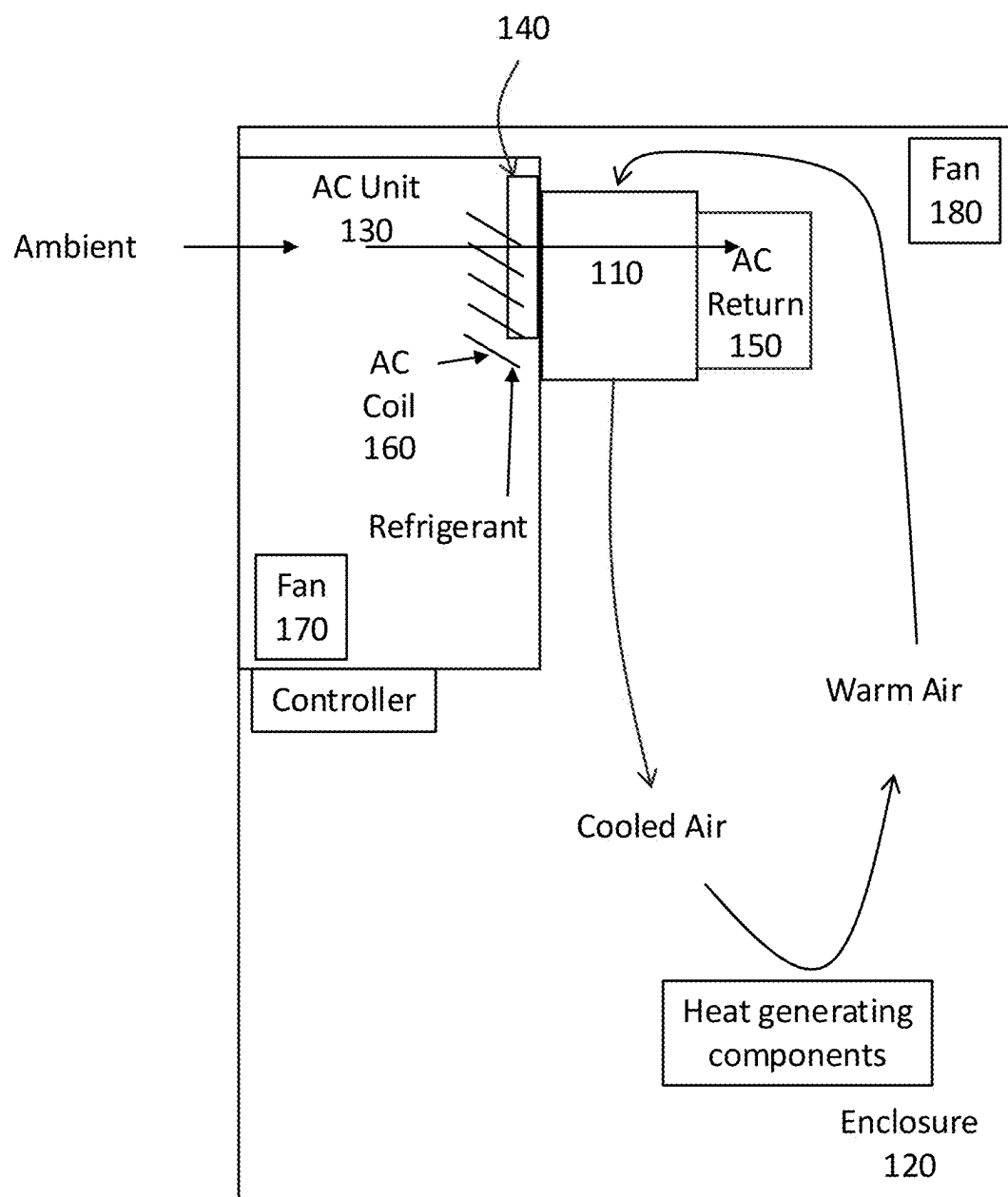


FIG. 1

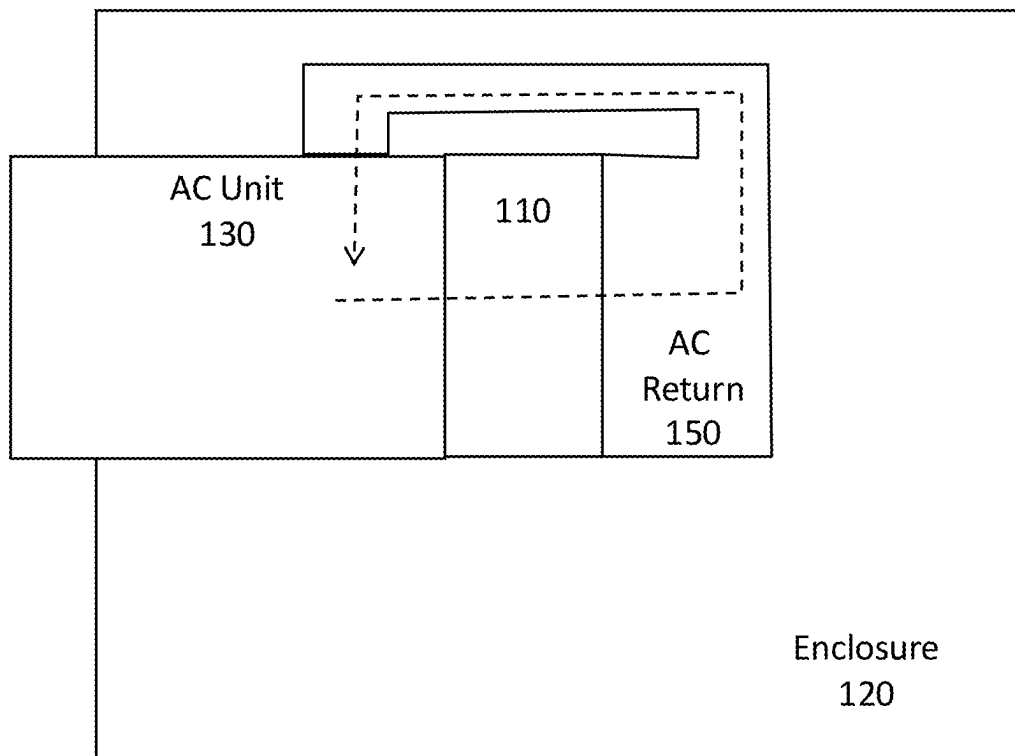


FIG. 1A

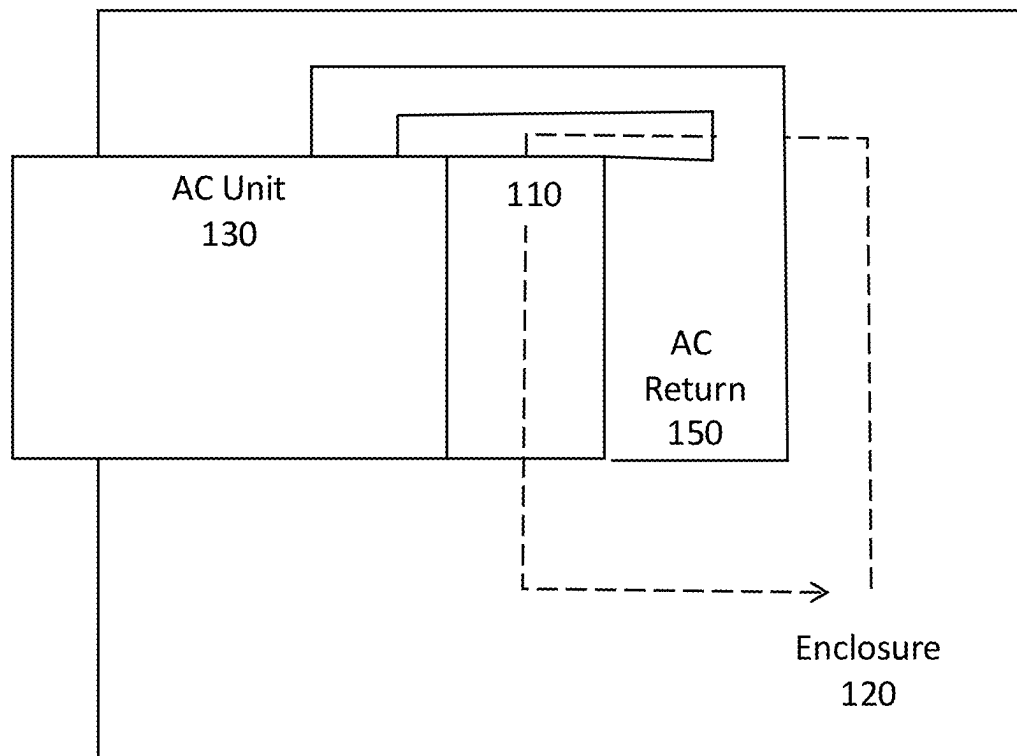
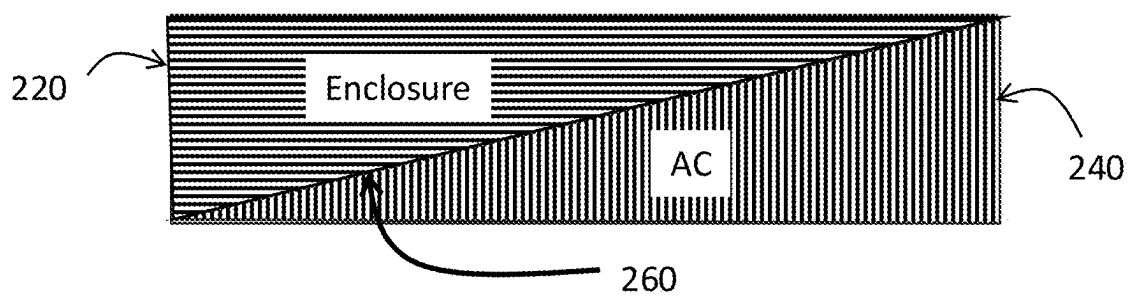
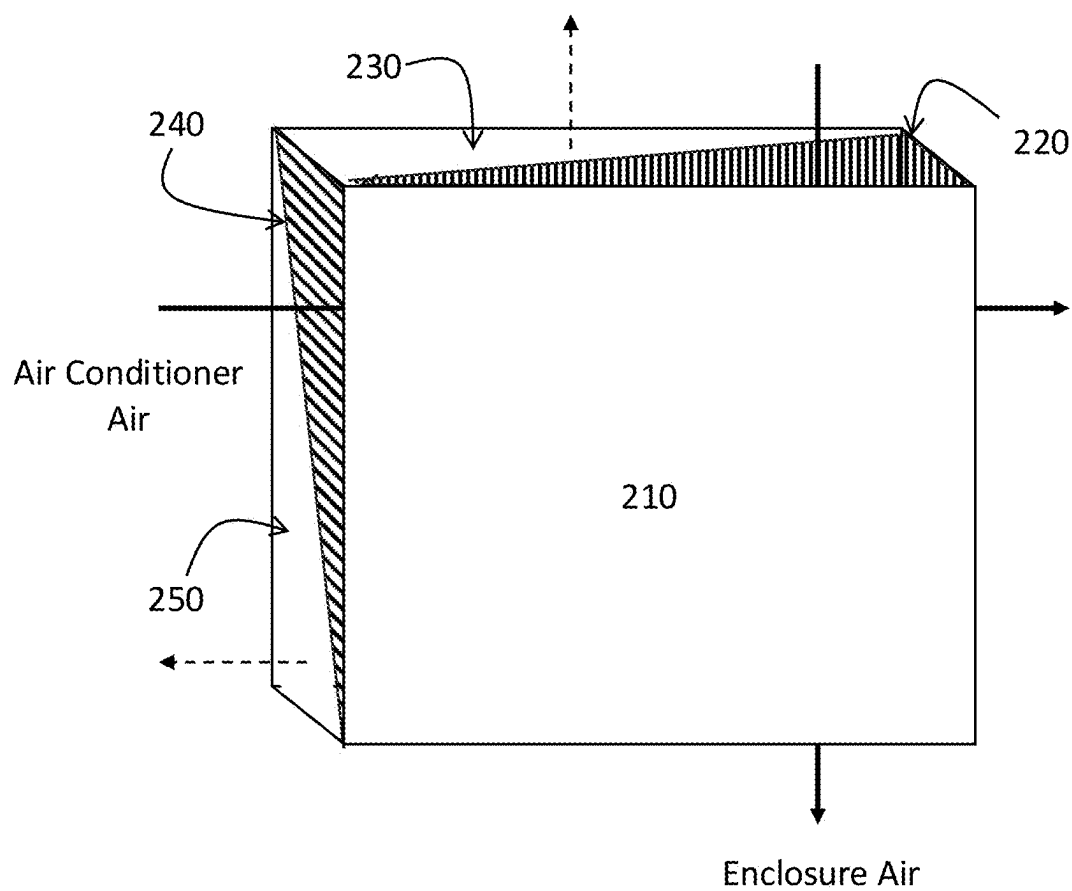


FIG. 1B



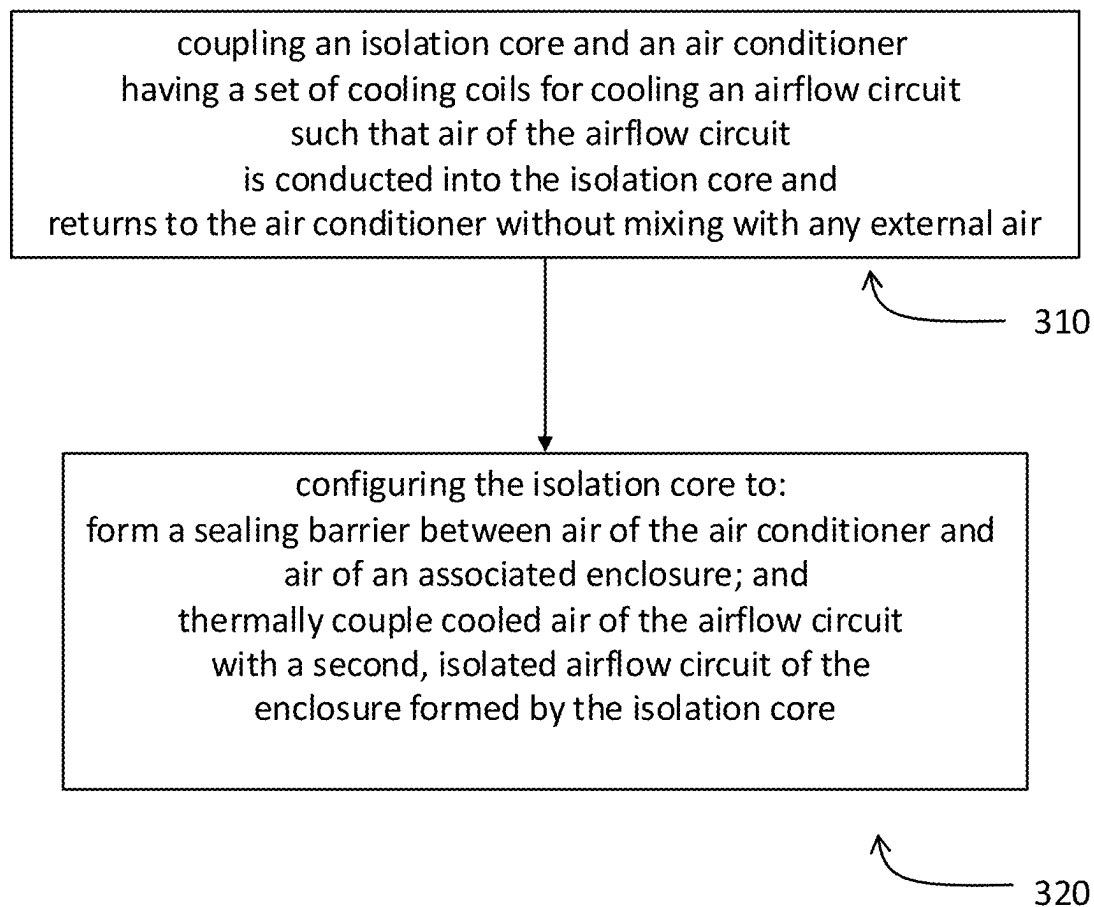


FIG. 3

REFRIGERANT ISOLATION FOR COOLING SOLUTIONS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to U.S. provisional patent application Ser. No. 63/562,511, filed Mar. 7, 2024, and having the same title, the content of which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] Various cooling equipment such as air conditioners, heat exchangers, in-line compressed air coolers and filtered fan systems are used to maintain required operating temperatures within an enclosure. The cooling solution(s) chosen may be dependent on the enclosure type, for example a sealed industrial enclosure housing heat generating components such as industrial equipment operating in a hazardous environment, servers, or other electronics, etc. In some cases, other enclosures are of interest, such as a house or dwelling or other non-hazardous environment.

BRIEF SUMMARY

[0003] An isolation core is used to isolate an air conditioning (AC) unit and an external environment such as an enclosure to be cooled. An isolation core may take the form of a cross-flow or counterflow heat exchanger which acts to thermally couple the flows of an AC unit and an enclosure air inlet and return without permitting the airflows to intermix. The isolation core thus facilitates cooling of enclosures while isolating flammable refrigerant from the enclosure air, requiring the refrigerant to remain within the AC unit even if an interior component of the AC unit such as a cooling coil, etc., leaks.

[0004] In summary, an embodiment provides a device, comprising: an isolation core configured to couple to an air conditioner, the isolation core further configured to: form a sealing barrier between air of the air conditioner and air of an associated enclosure; and thermally couple cooled air of an airflow circuit of the air conditioner with a second, isolated airflow circuit of the enclosure.

[0005] Another embodiment provides a system, comprising: an isolation core; and an air conditioner having a set of cooling coils for cooling an airflow circuit and conducting air of the airflow circuit towards the isolation core; the isolation core being configured to: couple to the air conditioner; form a sealing barrier between air of the air conditioner and air of an associated enclosure; and thermally couple cooled air of the airflow circuit of the air conditioner with a second, isolated airflow circuit of the enclosure.

[0006] A further embodiment provides a method, comprising: coupling an isolation core and an air conditioner having a set of cooling coils for cooling an airflow circuit such that air of the airflow circuit is conducted into the isolation core and returns to the air conditioner without mixing with any external air; and configuring the isolation core to: form a sealing barrier between air of the air conditioner and air of an associated enclosure; and thermally couple cooled air of the airflow circuit of the air conditioner with a second, isolated airflow circuit of the enclosure formed by the isolation core.

[0007] The foregoing is a summary and thus may contain simplifications, generalizations, and omissions of detail;

consequently, those skilled in the art will appreciate that the summary is illustrative only and is not intended to be in any way limiting.

[0008] For a better understanding of the embodiments, together with other and further features and advantages thereof, reference is made to the following description, taken in conjunction with the accompanying drawings. The scope of the invention will be pointed out in the appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 illustrates a side view of an example system according to an embodiment.

[0010] FIG. 1A illustrates a top view of an example system according to an embodiment.

[0011] FIG. 1B illustrates another top view of an example system according to an embodiment.

[0012] FIG. 2 illustrates an example device according to an embodiment.

[0013] FIG. 2A illustrates another example device according to an embodiment.

[0014] FIG. 3 illustrates a method according to an embodiment.

DETAILED DESCRIPTION

[0015] It will be readily understood that the components of the embodiments, as generally described and illustrated in the figures herein, may be arranged and designed in a wide variety of different configurations in addition to the described example embodiments. Thus, the following more detailed description of the example embodiments, as represented in the figures, is not intended to limit the scope of the claims, but is merely representative of those embodiments.

[0016] Reference throughout this specification to “embodiment(s)” (or the like) means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, appearances of the phrases “according to embodiments” or “in an embodiment” (or the like) in various places throughout this specification are not necessarily all referring to the same embodiment.

[0017] Furthermore, the described features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. In the following description, numerous specific details are provided to give a thorough understanding of example embodiments. One skilled in the relevant art will recognize, however, that aspects can be practiced without one or more of the specific details, or with other methods, components, materials, etc. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obfuscation.

[0018] In certain cases, an environment to be cooled may be hazardous, for example have flammable or explosive gasses in the atmosphere. In such cases, cooling systems may include various mechanisms to prevent ignition of the ambient environment, the enclosure environment, or a combination thereof. Conventional mechanisms have excluded from the potential sources of hazardous gasses in the cooling context the refrigerant(s) used in the cooling circuit of the air conditioner (AC) unit. This is because AC units typically utilize a refrigerant that is non-flammable.

[0019] However, given an emphasis on reducing environmental impact of cooling systems, interest has increased in using refrigerants that are thought to be less environmentally

harmful, for example having a smaller global warming impact or potential. In this respect, refrigerants may be restricted to certain types of substances. In many cases, the refrigerant substance preferred from an environmental perspective introduces difficulties with respect to operating cooling equipment in a safe manner because these refrigerants are mildly flammable (e.g., A2, A2L refrigerant types), and in some cases extremely flammable (e.g., A2 refrigerant type), introducing a new source of possible ignition to the cooling system.

[0020] By way of example, refrigerants are being promoted for environmental reasons that are flammable, for example propane (A3 refrigerant type) or an A2 type refrigerant such as A2L. This introduces a complexity in that even if a cooling system is made safe with respect to an ambient environment, the enclosure air, or a subcomponent of the AC unit, the cooling system refrigerant itself now provides a source of hazardous material, i.e., the flammable refrigerant may leak into an AC unit and the enclosure to which it is attached.

[0021] An embodiment therefore provides a cooling system including an AC unit and an isolation core. In an embodiment, the air conditioning unit and the isolation core are configured to be used in cooling an enclosure. Different types of enclosures may be cooled, for example an industrial enclosure, a server rack enclosure, a telecommunications cabinet, or a dwelling.

[0022] In an embodiment, the isolation core acts to isolate the fluid (e.g., gas) of the AC unit from the air/gas of the enclosure. In an embodiment, the isolation core is configured as a barrier to fluid (gas) communication between the air conditioner and the enclosure but permits heat exchange (thermal coupling) between the fluids (gasses) thereof. The thermal coupling may be facilitated by a physical element or elements, such as fins of a heat exchanger.

[0023] In an embodiment, the isolation core is a heat exchanger isolation core. The heat exchanger isolation core is configured as an interface between the AC unit air and the enclosure air (or airways to and from the enclosure) to be cooled.

[0024] In an embodiment, the heat exchanger isolation core is positioned to be in fluid (gas) communication with the cooling coils of the air conditioning unit. The heat exchanger is also positioned to be in communication with the enclosure air. In an embodiment, the respective parts of the heat exchanger isolation core forming conduits for the (air conditioner) airflow circuit and the enclosure airflow circuit are sealed with respect to one another, precluding the possibility that refrigerant present in the AC unit airflow circuit, e.g., leaked from a cooling coil, could enter the enclosure air. This permits flammable refrigerant to be used more safely.

[0025] In an embodiment, the heat exchanger isolation core is oriented to direct warm air to be cooled (from the enclosure) to a top of a cooling coil (warm part) of the AC unit, downward towards the bottom of the cooling coil (cool part) of the AC unit. This promotes cooling efficiency. In an embodiment, this orientation increases the efficiency of cooling by cooling the warm air of the enclosure entering the isolation core starting at the top (or warmest) portion of the cooling coil of the AC unit and transiting it in a downward fashion through the isolation core to the bottom, proximate to the coolest part of the cooling coil of the AC unit. As will be appreciated from review of the disclosure, this cooling

arrangement does not mean that air to be cooled (from the enclosure) is physically contacted with the cooling coil of the AC unit; rather, an indirect thermal coupling is used and the orientation of the isolation core conduits (e.g., heat exchanger) guides warm air along components of the isolation core, e.g., metal fins, such that it encounters those positioned to communicate heat or thermal energy proximate to the top portion of the cooling coil first, and likewise proceeding in a downward fashion to encounter lower portions of the isolation core, towards the bottom or coolest portion of the cooling coil, with a gradient in between.

[0026] In an embodiment, one or more dampers such as ECONOMIZER dampers of Ice Qube, Inc. of Greensburg Pennsylvania may be included. For example, by including dampers in the AC (evaporator) circuit, during lower ambient conditions (using some temperature threshold, such as ambient temperature below a certain temperature), the dampers can be opened and when the compressor/refrigerant is not needed, the system (isolation core) operates as a passive heat exchanger to remove heat in a passive mode without compromising the closed-looped nature of the system.

[0027] An embodiment includes a method of providing an isolation core for an air conditioner.

[0028] An embodiment includes a method of orienting the isolation core with respect to one or more components of the air conditioner to maximize its cooling efficiency.

[0029] In some embodiments, the air conditioning unit and the isolation core are configured to operate in a hazardous location or environment, whereas certain embodiments are configured for operating in a non-hazardous location or environment.

[0030] Referring to FIG. 1, an air conditioner (AC) unit 130 is configured with various components including a compressor, condenser, fan(s) 170, and cooling coils 160. In FIG. 1, one of the cooling coils 160 (AC evaporator coil) is drawn to illustrate a working example. In an embodiment, flammable refrigerant is cooled in AC unit 130 and provided to a bottom of the coil 160 (indicated by the arrow showing refrigerant entering the bottom), transiting upward through the coil 160 and providing heat absorption and cooling at a first inlet side of an isolation core 110 (left margin of isolation core 110, accepting AC airflow circuit). In an embodiment, the isolation core 110 is a heat exchanger (HX). In an embodiment, AC unit airflow circuit is provided such that cool air that enters the isolation core 110 after transiting over coils 160, providing cooling airflow to the fins of the HX and exiting as warmed airflow out the other side of isolation core 110 (via AC return 150 shown in FIG. 1). This permits the AC airflow circuit to be delivered to an AC return 150 in a closed loop manner, i.e., air or other fluids from AC unit 130 cannot exit isolation core 110 into the enclosure 120.

[0031] The enclosure 120 is supplied with an airflow, e.g., via fan(s) 180, that transits warm air through isolation core 110 from a top inlet and downward, oriented opposite the flow of refrigerant in coil 160 orientation, exiting out the bottom of isolation core 110 as cooled air return for enclosure 120. As described herein, orienting isolation core's 110 enclosure side inlet and return (top and bottom as shown in FIG. 1) as indicated in FIG. 1 acts to maximize the efficiency of the cooling system.

[0032] FIG. 1A shows a top view of the example system illustrated in FIG. 1. In FIG. 1A it can be appreciated that an

AC airflow circuit (dashed line) acts to exit AC unit **130**, transit through isolation core **110**, enter an AC return **150**, and circulate again to AC unit **130**. In similar fashion, FIG. 1B provides a top view of the example system of FIG. 1 highlighting the enclosure airflow circuit (dashed line) showing that it is circulated into and out of isolation core **110** without interacting with AC unit **130**. Further, as described herein, each of the AC airflow circuit (dashed line of FIG. 1A) and the enclosure airflow circuit (dashed line of FIG. 1B) are sealed from one another via isolation core **110**, i.e., a flammable refrigerant such as propane present in AC airflow circuit cannot mix with enclosure airflow circuit.

[0033] Referring to FIG. 2, isolation core **210** isolates fluids of AC unit **130** from fluids of the enclosure **120** and prevents mixing. This allows for use of flammable refrigerant, as any leakage, e.g., from coils **160**, etc., of the AC unit **130** cannot escape the AC unit **130** itself and enter into the enclosure **120**. An example of a heat exchanger as an isolation core **210** is a convoluted core heat exchanger, for example a cross-flow heat exchanger or a counterflow heat exchanger available from Ice Qube, Inc. of Greensburg Pennsylvania under the names XF SERIES or CF SERIES heat exchangers. As will be understood, a convoluted heat exchanger as illustrated in FIG. 2 and FIG. 2A directs two different and isolated (fluid isolation) airflows, allowing heat exchange there-between (thermally communicating).

[0034] As shown in FIG. 2, the isolation core **210** may feature an inlet and paired outlet for each of the respective AC airflow circuit and the enclosure airflow circuit. As illustrated in FIG. 2, the AC airflow circuit transits into **240** and exits the opposite (right) side, whereas the enclosure airflow circuit enters at **220** and exits at the opposite (bottom) side). The configuration in FIG. 2 may be referred to as a cross-flow configuration, where the AC airflow circuit and the enclosure airflow circuit have paths that form a cross in the isolation core **210**, e.g., at about 90 degrees. As illustrated, barriers **230**, **250** preclude the airflow circuits from combining. That is, the AC and enclosure airflow circuits are not permitted to intermix.

[0035] As illustrated in FIG. 2A, a top view or side-on view of a counter flow heat exchanger isolation core **210**, may also take the form of a counterflow heat exchanger that passes the AC airflow circuit and the enclosure airflow circuit in parallel to one another, e.g., in opposing directions. In such a case, the AC airflow circuit enters in **240** (downward into top view of FIG. 2A) and exits the opposite side (below), whereas the enclosure airflow circuit enters in **220** at the opposite side (below) and exits out the top. As illustrated in FIG. 2A, an internal barrier element **260** prevents the AC airflow circuit and the enclosure airflow circuit from mixing. As may be appreciated, appropriate ducting for inlets and exits of isolation core chosen **110**, **210** permit fully isolating the AC airflow circuit and the enclosure airflow circuit.

[0036] Referring to FIG. 3, an embodiment provides a method, including coupling an isolation core and an air conditioner having a set of cooling coils for cooling an AC airflow circuit such that air of the AC airflow circuit is conducted into the isolation core and returns to the air conditioner without mixing with any external air, indicated at **310**. The method may further include configuring the isolation core to: form a sealing barrier between air of the air conditioner and air of an associated enclosure; and thermally couple cooled air of the (AC) airflow circuit with a second,

isolated airflow circuit of the enclosure formed by the isolation core, indicated at **320**. An embodiment may include a product by process as described throughout, e.g., an isolation core, AC unit, system, component, or combination thereof formed by performing one or more acts described herein.

[0037] Referring back to FIG. 1, an embodiment may include one or more dampers **140** (passive, e.g., gravity and/or airflow powered, or active/motorized, e.g., using electric motors). For example, for dampers **140** in the AC evaporator circuit, during lower temperature ambient conditions (as determined by some temperature threshold or set point), the dampers **140** can be opened and when the compressor/refrigerant is not needed, the system operates as a passive heat exchanger (e.g., using isolation core **110** to exchange ambient air for AC unit **130** cooled air) to remove heat in a “passive” mode without compromising the closed-looped nature of the enclosure air circuit. In a passive mode, the AC unit **130** is not operating to produce cooling via coil **160**. As may be appreciated, this may require appropriate ducting or components acting to establish an airflow between the isolation core **110** and the ambient environment, e.g., via AC unit **130** as shown in FIG. 1. This permits ambient air to circulate through the isolation core **110** via the AC unit **130**, as indicated by two arrows transiting AC unit **130** to isolation core **110**, cooling in a passive mode while maintaining the closed-loop circuit.

[0038] If dampers **140** are added to the evaporator coil (**160**) air circuit, in lower temperature ambient conditions the isolation core **110** may become a passive heat exchanger and continue to remove heat from the enclosure without compromising the closed loop enclosure air circuit. This application is a valuable addition and adds a level of low-cost cooling in one unit. It is a good application for ordinary locations and even more so in hazardous locations. Dampers **140** conventionally can't be used in hazardous (e.g., gas and oil) applications where explosive gasses exist in the ambient air because gasses must be kept out of the enclosure **120** but with the isolation core **110** arrangement, a closed loop environment will be maintained in the enclosure **120** and still use cooler ambient air to remove heat.

[0039] Referring again to FIG. 1, an embodiment may include a controller, which acts to control AC unit **130**, dampers **140** (which may be only a single damper), etc. In an embodiment, an example device that may be used in implementing one or more embodiments of a controller may be provided in the form of a microcontroller computing device or a control panel that operates one or more components, for example of the AC unit **130**.

[0040] Controller may execute program instructions or code or operate using dedicated circuitry configured to process data or signals and perform other functionality of the embodiments. Components of controller may include, but are not limited to, a processing unit, which may take a variety of forms such as a central processing unit (CPU), a programmable circuit or other programmable hardware, non-programmable hardware, a combination of the foregoing, etc., a system memory controller and memory, as well as a system bus that couples various system components including the system memory to the processing unit. It is noted that in certain implementations, controller may take a reduced or simplified form, such as a micro-control unit implemented in a control panel of a cooling system, or even non-programmable hardware such as a series of relays,

switches, or circuits, where certain of the components of controller are omitted or combined, or the “controller” is formed by one or more of these elements.

[0041] Controller may include or have access to a variety of non-transitory computer readable media such as memory. Memory may include non-transitory computer readable storage media in the form of volatile and/or nonvolatile memory devices such as read only memory (ROM) and/or random-access memory (RAM). By way of example, and not limitation, memory may also include an operating system, application programs, other program modules, and program data. For example, memory may include application programs such as variable speed control software and/or air conditioner operational software for implementing various cooling and/or heating protocols, as described herein. Data may be transmitted by wired or wireless communication elements, respectively, e.g., to or from first device to another device, e.g., communication between a remote device or system such as controller.

[0042] A user can interface with (for example, enter commands and information) the controller through input devices such as a touch screen, keypad, etc. A monitor or other type of display screen or device may also be connected to system bus via an interface. Controller may operate in a networked or distributed environment using logical connections to one or more other remote computers or databases. The logical connections may include a network, such local area network (LAN) or a wide area network (WAN) but may also include other networks/buses. In one example, controller is remotely controllable via Ethernet.

[0043] It should be noted that various functions described herein may be implemented using processor executable instructions stored on a non-transitory storage medium or device or using dedicated circuitry or circuits. A non-transitory storage device may be, for example, an electronic, electromagnetic, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples of a non-transitory storage medium include the following: a portable computer diskette, a hard disk, a random-access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a portable compact disc read-only memory (CD-ROM), a solid-state drive, or any suitable combination of the foregoing. In the context of this document “non-transitory” media includes all media except non-statutory signal media.

[0044] Program code embodied on a non-transitory storage medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, RF, etc., or any suitable combination of the foregoing.

[0045] Program code for carrying out operations may be written in any combination of one or more programming languages. The program code may execute entirely on a single device, partly on a single device, as a stand-alone software package, partly on single device and partly on another device, or entirely on the other device. In some cases, the devices may be connected through any type of connection or network, including a local area network (LAN) or a wide area network (WAN), a personal area network (PAN) or the connection may be made through other devices (for example, through the Internet using an Internet Service Provider), through wireless connections, or

through a hard wire connection, such as over a USB or another power and data connection.

[0046] It is worth noting that while specific elements are illustrated in the figures, and a particular ordering or organization of elements or steps has been illustrated, these are non-limiting examples. In certain contexts, two or more elements or steps may be combined into an equivalent element or step, an element or step may be split into two or more equivalent elements or steps, or certain elements or steps may be re-ordered or re-organized or omitted as appropriate, as the explicit illustrated examples are used only for descriptive purposes and are not to be construed as limiting.

[0047] If used herein, the term “about” shall include ordinary rounding from a base number to a nearest significant digit, for example a base number of 10.1 may be rounded to 10.0 and 10.2.

[0048] As used herein, the singular “a” and “an” may be construed as including the plural “one or more” unless clearly indicated otherwise. Likewise, plural reference may be consolidated where appropriate.

[0049] This disclosure has been presented for purposes of illustration and description but is not intended to be exhaustive or limiting. Many modifications and variations will be apparent to those of ordinary skill in the art. The example embodiments were chosen and described in order to explain principles and practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

[0050] Thus, although illustrative example embodiments have been described herein with reference to the accompanying figures, it is to be understood that this description is not limiting and that various other changes and modifications may be affected by one skilled in the art without departing from the scope or spirit of the disclosure.

What is claimed is:

1. A device, comprising:
 - an isolation core configured to couple to an air conditioner, the isolation core further configured to:
 - form a sealing barrier between air of the air conditioner and air of an associated enclosure; and
 - thermally couple cooled air of an airflow circuit with a second, isolated airflow circuit of the enclosure.
 2. The devices of claim 1, wherein each of the airflow circuit of the air conditioner and the second, isolated airflow circuit of the enclosure transit through the isolation core.
 3. The device of claim 1, wherein the isolation core comprises an airflow return configured to return the airflow circuit of the air conditioner back to the air conditioner.
 4. The device of claim 1, where the isolation core comprises:
 - a first set of airway conduits configured to form part of the airflow circuit of the air conditioner; and
 - a second set of airway conduits configured to form part of the second, isolated airflow circuit of the enclosure.
 5. The device of claim 1, wherein the isolation core is a heat exchanger.
 6. The device of claim 5, wherein the heat exchanger is a cross-flow heat exchanger.
 7. The device of claim 5, wherein the heat exchanger is a counterflow heat exchanger.
 8. The device of claim 1, comprising a damper configured to operate responsive to a temperature threshold.

9. The device of claim 8, wherein the damper opens to ambient in response to the temperature threshold.

10. The method of claim 8, wherein the damper acts to close the airflow circuit of the air conditioner and open an ambient airflow circuit.

11. A system, comprising:

an isolation core; and

an air conditioner having a set of cooling coils for cooling an airflow circuit and conducting air of the airflow circuit towards the isolation core;

the isolation core being configured to:

couple to the air conditioner;

form a sealing barrier between air of the air conditioner and air of an associated enclosure; and

thermally couple cooled air of the airflow circuit with a second, isolated airflow circuit of the enclosure.

12. The system of claim 11, wherein each of the airflow circuit of the air conditioner and the second, isolated airflow circuit of the enclosure transit through the isolation core.

13. The system of claim 11, comprising an airflow return configured to return the airflow circuit of the air conditioner back to the air conditioner.

14. The system of claim 11, where the isolation core comprises:

a first set of airway conduits configured to form part of the airflow circuit of the air conditioner; and

a second set of airway conduits configured to form part of the second, isolated airflow circuit of the enclosure.

15. The system of claim 11, wherein the isolation core is a heat exchanger.

16. The system of claim 15, wherein the heat exchanger is a cross-flow heat exchanger.

17. The system of claim 15, wherein the heat exchanger is a counterflow heat exchanger.

18. The system of claim 11, comprising a damper configured to operate responsive to a temperature threshold.

19. The system of claim 18, wherein the damper opens to ambient in response to the temperature threshold.

20. A method, comprising:

coupling an isolation core and an air conditioner having a set of cooling coils for cooling an airflow circuit such that air of the air conditioner is conducted into the isolation core and returns to the air conditioner without mixing with any external air; and

configuring the isolation core to:

form a sealing barrier between air of the air conditioner and air of an associated enclosure; and

thermally couple cooled air of the airflow circuit with a second, isolated airflow circuit of the enclosure formed by the isolation core.

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